Arjun Krishna
2467 03428
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DBMS ASSIGNMENT - 3

Question 5.11

a. Insert <'Robert', 'F', 'Scott', '943775543', '1972-06-21', '2365 Newcastle Rd', Bellaire, TX', M, 58000, '888665555', 1> into EMPLOYEE.

There are no violations. The new primary key Ssn ('943775543') is unique. The foreign key Super_ssn ('888665555') correctly references an existing employee (James Borg). The foreign key Dno (1) correctly references an existing department (Headquarters).
Enforcement: The operation is accepted.

b. Insert <'ProductA', 4, 'Bellaire', 2> into PROJECT.

There is a Referential Integrity violations. The Dnum attribute acts as a foreign key referencing the DEPARTMENT table. There is no department and Dnumber = 2 in the current state (only 1, 4, and 5 exist).
Enforcement: The insertion operation is rejected.

c. Insert <'Production', 4, '943775543', '2007-10-01'> into DEPARTMENT.

There is an Entity Integrity (key Constraint) violations. The primary key Dnumber must be unique. Dnumber = 4 already exists for the 'Administration' department.
Enforcement: The insertion operation is rejected.

d. Insert <'677678989', NULL, '40.0'> into WORKS_ON.

There is an Entity Integrity violation. Pno is part of the Primary key (Essn, Pno) for the WORKS_ON table. Parts of a primary key cannot be NULL.

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There is also an Referential Integrity violation. The Essn ('677678989') does not exist in the EMPLOYEE table.

Enforcement: the Insertion operation is rejected.

e. Insert <'453453453', 'John', 'M', '1990-12-12', 'Spouse'> into DEPENDENT.

There are no violations. The Essn ('453453453') exists in the EMPLOYEE table (Joyce English). The combination of Essn and Dependent_name ('John') form a unique primary key for this table.

Enforcement: This operation is accepted.

f. Delete the WORKS_ON tuples with Essn = '333445555'.

There are no violations. No other tables contains a foreign key referencing the WORKS_ON table, so deleting records from it does not cause orphan records elsewhere.

Enforcement: This operation is accepted. The four tuples corresponding to Franklin Wong will be ~~[illegible]~~ removed.

g. Delete the EMPLOYEE tuple with SSn = '987654321'.

There is a Referential Integrity. Jennifer Wallace's Ssn ('987654321') is referenced in several other places:

- As Super_Ssn in the EMPLOYEE table (supervising Alicia Zelaya and Ahmad Jabbar).
- As Mgr_ssn in the DEPARTMENT table (managing department 4).
- As Essn in the WORKS_ON table.
- As Essn in the DEPENDENT table (Abner).

Enforcement:

- 1st option → Reject the operation.
- 2nd option → Cascade the deletion to WORKS_ON and DEPENDENT. Set NULL for Super_ssn in EMPLOYEE and Mgr_ssn in DEPARTMENT (assuming these attributes allow NULL values).

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Aryan Kushwaha
245703428
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h. Delete the PROJECT tuple with Pname = 'ProductX'.

There is a Referential Integrity violation. The Primary Key for 'ProductX' is Pnumber = 1. The value is referenced as Pno in the WORKS_ON table by multiple employees.

Enforcement:

- 1st option → Reject the deletion.
- 2st option → Cascade the deletion, which would automatically delete all related tuples in the WORKS_ON table where Pno = 1. (Setting to NULL is not possible since Pno is part of the Primary Key in WORKS_ON).

i. Modify the Mgr_ssn and Mgr_start_date of the DEPARTMENT tuple with Dnumber = 5 to '123456789' and '2007-10-01', respectively.

There are no violations. The new Mgr_SSN('123456789') correctly references an existing employee (John Smith).

Enforcement: the operation is accepted.

j. Modify the Super_ssn attribute of the EMPLOYEE tuple with SSN = '999887777' to '943775543'.

There is a Referential Integrity. The new Super_SSN value ('943775543') does not exist as an SSN in the EMPLOYEE table in the current database state.

Enforcement: Reject the modification.

k. Modify the Hours attribute of the WORKS_ON tuple with Essn = '999887777' and Pno = 10 to '5.0'.

There are no violations. The specified tuple exists, and modifying a non-Key attribute like Hours to a valid decimal values does not violate relational constraints.

Enforcement: The operation is accepted.

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problem 5.14]

To maintain referential integrity between the tables, we can define the following foreign keys.

Assumptions made for these definitions:

- The primary key for each table is the attribute ending in #. (e.g., Cust# for CUSTOMER; Order# for ORDER).
- For linking tables like ORDER_ITEM and SHIPMENT, the primary key is a composite of the foreign keys (e.g., Order# and Item# for ORDER_ITEM; Order#, Warehouse#, and potentially Ship_date for SHIPMENT, depending on whether multiple shipments for the same order from the same warehouse on different dates are allowed).

Defined Foreign Keys:

- In the ORDER relation:
  - Cust# is a foreign key referencing the Cust# primary key in the CUSTOMER relation.
  - An order must be placed by a valid, existing customer.
- In the ORER_ITEM relation:
  - Order# is a foreign key referencing the Order# primary key in the ORDERS relation.
    (An item being ordered must belong to a valid, existing order)
  - Item# is a foreign key referencing the Item# primary key in the ITEM relation.
    (The item listed in an order must be a valid product in the company catalog.)
- In the SHIPMENT relation:
  - Order# is a foreign key referencing the Warehouse# primary key in the WAREHOUSE relation.
    (The order must be shipped from a valid, existing warehouse location).

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Aryan Kushwaha
245703428
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Beyond primary and foreign keys, several other constraints should be enforced to ensure data quality and business logic accuracy.

Domain and Data type Constraints:

- Key attributes like Cname (Customer Name), Odate (Order date), and Unit_price should not be allowed to be NULL.
- Ord_amt (Total Order Amount) must be > 0.
- Qty (Quantity ordered) must be an Integer >0.
- Unit_price must be >0.

Semantic (Business Rule) Constraints:

- The Ship_date in the SHIPMENT relation must be ≥ the Odate in the corresponding ORDER relation. An order cannot be shipped before it is placed.
- The Ord_amt in the ORDER table is a derived attribute. A constraint (often implemented via database triggers) should ensure that for any given Order#, the Ord_amt exactly equals the sum of (Qty × Unit_price) for all corresponding records in the ORDER_ITEM and ITEM tables.

$$Ord_amt = \sum (Qty \times Unit_price) \text{ for a specific Order\#.}$$

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Problem 5.15]

To define foreign Keys correctly, we must first assume the primary Keys for each relation based on the context:

- SALESPERSON: Ssn is the primary Key, uniquely identifying each salesperson.
- TRIP: Trip_id is the primary Key, uniquely identifying each specific business trip.
- EXPENSE: The problem states that "a trip can be charged to one or more accounts." Because of this many-to-many relationship between trips and accounts for expenses, neither Trip_id nor Account# can be a primary key on its own. The primary key must be the composite of (Trip_id, Account#).

Based on the assumed primary keys, the following foreign keys should be specified to maintain Referential Integrity:

- In the TRIP relation:

Ssn is a foreign key referencing the Ssn primary key in the SALESPERSON relation, because every business trip must be taken by a valid, existing salesperson recorded in the system.

- In the EXPENSE relation:

Trip_id is a foreign key referencing the Trip_id primary key in the TRIP relation, because every expense amount recorded must be associated with a valid, previously logged business trip.

While not explicitly defined in the short list of relations provided, in a complete database schema, you would typically also have:

- In SALESPERSON: Dept_no would act as a foreign key referencing the primary key of a DEPARTMENT table (e.g., DEPARTMENT (Dept_no, Dept_name)).
- In EXPENSE: Account# would act a foreign key referencing the primary key of an ACCOUNT table (e.g., ACCOUNT (Account#, Account_name)).

Mayank Kushwaha
245703428
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problem 5.16

To define foreign keys correctly, we must first establish the primary keys for each relation. Based on standard database design principles, we can assume the following:

- Ssn (STUDENT) is the primary key, uniquely identifying each student.
- Course# (COURSE) is the primary key, uniquely identifying each course.
- Book_Isbn (TEXT) is the primary key, uniquely identifying each textbook
- (Ssn, Course#, Quarter) (ENROL) is a composite key, a student can take a specific course in a specific quarter.
- (Course#, Quarter, Book_Isbn) (BOOK_ADOPTION) is a composite key, a single course might require multiple books in a given quarter, and the same book might be used in different courses or different quarters.

Based on the assumed primary keys, the following foreign keys should be specified to maintain referential integrity:

- In the ENROLL relation:
  - Ssn is a foreign key referencing the Ssn primary key in the STUDENT relation.
    (Every enrollment record must be associated with a valid, existing student in the database).
  - Course# is a foreign key referencing the Course# primary key in the COURSE relation.
    (A student can only be enrolled in a valid, existing course offered by the institution).
- In the BOOK_ADOPTION relation:
  - Course# is a foreign key referencing the Course# primary key in the COURSE relation.
    (A book adoption record must be tied to a valid, existing course).
  - Book_Isbn is a foreign key referencing the Book_Isbn primary key in the TEXT relation.
    (Any book adopted for a course must exist in the main catalog of textbooks.)

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While not explicitly defined in the short list of relations provided, a fully normalized schema might also include a generic DEPARTMENT table. If that were the case:

- In STUDENT, the Major attribute could act as a foreign key referencing the DEPARTMENT table.
- In COURSE, the Dept attribute could also act as a foreign key referencing the DEPARTMENT table.

question 5.17

To define foreign keys correctly, we must first establish the primary key for each relation based on the given context:

- CAR: Serial_no is the primary key, uniquely identifying each automobile.
- OPTION: Since a car can have multiple option like a sunroof, leather seats, etc, a single option name or serial number is not unique. The primary key must be the composite of (Serial_no, Option_name).
- SALE: A specific car is usually sold once in this context, but a salesperson can sell many cars. The Serial_no is the most logical unique identifier for a single sale transaction. Therefore, the primary key is Serial_no.
- SALESPERSON: Salesperson_id is the primary key, uniquely identifying each employee.

Based on the assumed primary keys, the following foreign key should be specified to maintain referential integrity:

- In the OPTION relation:
  - Serial_no is a foreign key referencing the Serial_no primary key in the CAR relation.

(Any optional equipment listed must be installed on a valid, existing car in the inventory).

Aryan Kushwaha
245703428
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- In the SALE relation:
  - Serial_no is a foreign key referencing the Serial_no primary key in the CAR relation. (a recorded sale must correspond to a valid, existing car in the dealership's inventory).
  - Salesperson_id is a foreign key referencing the Salesperson_id primary key in the SALESPERSON relation. (A recorded sales must be attributed to a valid, existing salesperson employed by the dealership.

Let's populate the table with a few sample tuples to demonstrate the relationships

CAR			
Serial_no	Model	Manufacturer	Price
VIN12345	Civic	Honda	25000
VIN67890	Camry	Toyota	28000
VIN11223	Mustang	Ford	38000

OPTION		
Serial_no	Option_name	Price
VIN12345	Sunroof	1000
VIN12345	Alloy Wheels	800
VIN67890	Leather Seats	1500

SALESPERSON		
Salesperson_id	Name	Phone
SP001	Jane Doe	555-0101
SP002	John Smith	555-0102

SALE			
Salesperson_id	Serial_no	Date	Sale_price
SP001	VIN12345	2024-05-10	26800
SP002	VIN67890	2024-05-12	29000

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Examples of Insertions : Violating Referential Integrity (Invalid):

- Invalid SALE Insertion:

  INSERT INTO SALE VALUES ('SP003', 'VIN11223', '2024-05-15', 34000);

  (Referential integrity is violated because the Salesperson-id 'SP003' does not exist in the SALESPERSON relation.)

- Invalid OPTION Insertion:

  INSERT INTO OPTION VALUES ('VIN99999', 'Navigation', 1200);

  (Referential integrity is violated because the Serial-no 'VIN99999' does not exist in the CAR relation.)

Examples of Insertions : Maintaining Referential Integrity (Valid)

- Valid SALESPERSON Insertion:

  INSERT INTO SALESPERSON VALUES ('SP003', 'Emily Chen', '555-0103');

  (This is simply adds a new valid salesperson).

- Valid SALE Insertion (after adding Emily):

  INSERT INTO SALE VALUES ('SP003', 'VIN11223', '2024-05-15', 34000);

  ('SP003' now exists in SALESPERSON, and 'VIN11223' exists in the CAR table)